

CODE :-

```
import numpy as np
import numpy.random as ar
import matplotlib.pyplot as plt

X = np.array([[0,0],[0,1],[1,0],[1,1]])
t = np.array([0,0,0,1])
w = ar.rand(2)
print("w=",w)
b = ar.rand(1)
print("b=",b)
lr = ar.rand(1)
print("lr=",lr)

for epoch in range(50):
    print("epoch:",epoch)
    e=0
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
        yin = 1 if z >= 0 else 0
        error = t[i] - yin
        if(error!=0):
            e+=1
            print("error=",error)
            w += lr * error * X[i]
            b += lr * error
    if e==0:
        break;
print("Final weights:", w)
print("Final bias:", b)

yin = []
for i in range(len(X)):
    z = np.dot(X[i], w) + b
    yin.append(1 if z >= 0 else 0)

if np.array_equal(yin, t):
    print("Model is working for all inputs")
else:
    print("Model is not fully correct")
    print("Predictions:", predictions)
    print("Targets:", t)

mp.figure(figsize=(6,6))
X1=X[:,0]
X2=X[:,1]
mp.scatter(X1,X2,c=t,cmap="coolwarm",edgecolors="k", s=50)
x1=np.linspace(-7.5,10,100)
x2=-(x1*w[0]+b)/w[1]
mp.plot(x1,x2,c="black",label="Decision boundary")
plt.xlabel("x1")
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plt.ylabel("x2")
plt.title("Perceptron Decision Boundary for AND Gate")
plt.legend()
plt.grid()
```

OUTPUT :-

w= [0.67868496 0.67039307]

b= [0.36130979]

lr= [0.05547966]

epoch: 0

error= -1

error= -1

error= -1

epoch: 1

error= -1

error= -1

error= -1

epoch: 2

error= -1

error= -1

error= -1

epoch: 3

error= -1

error= -1

epoch: 4

error= -1

error= -1

epoch: 5

error= -1

epoch: 6

Final weights: [0.40128665 0.3375151]

Final bias: [-0.41540546]

Model is working for all inputs

